

Reddit Signal Radar - System Overview for Hypericum

What This System Does

The Reddit Signal Radar (RSR) is a custom intelligence tool that monitors Reddit in real time, identifies posts and threads where someone is describing a problem that Hypericum directly solves, and produces a ready-to-post comment draft for each one.

The end output - for every qualifying Reddit post - is:

1. A structured breakdown of the person's pain point, context, and what they want
2. A relevance assessment: how closely the post maps to Hypericum's expertise
3. A full comment draft, written in the voice of a knowledgeable practitioner, ready for human review and posting

The system does not post automatically. Every draft is reviewed before anything goes live. The goal is to surface the right conversations at the right moment and do 90% of the drafting work - the human makes the final call.

How It Works - All Layers

Layer 1: Event Ingestion

The system is installed on Reddit subreddits as a moderation app (built on Reddit's developer platform, Devvit). Every time someone submits a post or comment in a monitored subreddit, the system receives it instantly - no polling, no delay. This covers:

- New posts
- New comments
- Edited posts (updates the existing record)

The subreddits monitored are configured at setup time. Typical targets: r/dataengineering, r/MachineLearning, r/SaaS, r/entrepreneur, r/devops, r/BusinessIntelligence, r/dataanalysis.

Layer 2: Preprocessing and Noise Filter

Every piece of content is cleaned before analysis:

- URLs, markdown formatting, and code blocks are stripped

- The text is lowercased and normalised
- Known spam phrases are blocked (click here, dm me, etc.)
- Content shorter than 40 characters or fewer than 5 meaningful words is discarded

This ensures only real, substantive posts reach the analysis stage.

Layer 3: Intent Scoring

The cleaned text is scanned against a rule set of **150+ phrases** tuned to detect genuine pain, frustration, and active problem-solving. Each phrase carries a point value and an intent type.

Intent types:

Type	What it catches
frustration	Someone is at their limit with a current situation
complaint	Active dissatisfaction with a tool, process, or outcome
switching	Looking for alternatives, migrating away from something
discovery	Actively seeking a solution, asking for recommendations

Example phrases that score highly:

Phrase	Points	Why it matters
"works in demo but"	35	Classic AI production failure signal
"inconsistent classifications"	35	Core Hypericum problem domain
"reconciling reports"	35	Analytics reconciliation domain
"taxonomy mismatch"	35	Post-acquisition integration domain
"works for some clients"	32	Multi-tenant SaaS AI domain
"different definitions"	32	Analytics reconciliation domain
"auditable classification"	32	Regulatory audit domain
"inconsistent in production"	32	AI production failure domain
"breaking point"	35	High-urgency frustration signal

Points are additive. A post matching five phrases scores the sum. The score is capped at 100.

Score thresholds:

Score	Level	Action
0-29	Too low	Discarded
30-69	Medium	Saved, LLM insight extracted
70-100	High	Saved, LLM insight extracted, comment draft generated

Layer 4: Domain Cluster Assignment

Every qualifying post is assigned to one or more topic clusters based on keyword matching. The clusters are now aligned with Hypericum's six core problem domains:

Cluster	What it catches
ai-production-failure	RAG pipelines, LLM classification, hallucination, inconsistent AI outputs
analytics-reconciliation	Report discrepancies, metric definitions, "trust the data" problems
acquisition-integration	Incompatible taxonomies post-merger, product hierarchy mismatches
multitenant-saas-ai	SaaS vendors building AI across clients with different data models
regulatory-audit	Consumer Duty, GDPR, EU AI Act, audit trail, classification history
knowledge-graph-governance	Neo4j, ontology drift, taxonomy inconsistency, semantic layer
manual-workflow	Generic manual process pain (catch-all)
integration-gaps	Pipelines, connectors, data sync issues

A single post can match multiple clusters simultaneously.

Layer 5A: Signal Storage

Every qualifying signal is stored in Redis with:

- The full post content
- Intent score, level, and matched phrases
- Assigned clusters
- Engagement metrics (upvotes, comment count)
- A direct permalink back to the Reddit post

Signals are retained for 90 days. The system maintains up to 500 signals per query.

Layer 5B: LLM Insight Extraction (First AI Call)

For every signal that scores 30 or above, the system makes a call to **Gemini 2.5 Flash** to extract structured intelligence from the post. This runs asynchronously - it does not slow down the ingestion pipeline.

What is extracted per post:

Field	Description
Pain point	The core problem in one clear sentence
User context	Who this person is and their situation
Current workaround	How they are solving the problem today
Desired solution	What they actually want
Emotional tone	frustrated / annoyed / curious / desperate / hopeful / neutral
Urgency	high / medium / low
Marketing hook	A one-line hook capturing this person's specific need

This insight is stored alongside the signal and used as input for the next stage.

Layer 5C: Comment Draft Generation (Second AI Call)

For every signal that has an insight, the system makes a second call to **Gemini 2.5 Flash** - this time with the full Hypericum briefing embedded as context.

The model is instructed to:

1. Assess how closely the post maps to one of Hypericum's six problem domains
2. Determine whether Hypericum should be mentioned, and how prominently
3. Draft a full comment in plain prose, written as a knowledgeable practitioner - not as a product advertiser
4. Flag anything a human reviewer should adjust before posting

What the draft includes:

Field	Description
Relevance	direct - Hypericum is the natural answer; partial - Hypericum is one useful option; none - not relevant, comment answers the question without mentioning Hypericum
Relevance reason	One sentence explaining the relevance assessment
Domain match	Which of the 6 Hypericum problem domains this post maps to
Draft	The full comment text, ready for review and posting
Posting guidance	

Field	Description
	Notes for the human reviewer - suggested edits, tone caveats, things to double-check

Rules the AI follows when drafting:

- Mention Hypericum only where it is the natural answer to an explicitly described problem
- Never use promotional language or sales copy
- Never explain internal mechanics of how Hypericum works (patent pending protection applied)
- Write as a practitioner in a real technical conversation, not as a marketer
- Lead with the genuine answer to the question; Hypericum is cited where it fits, not inserted
- If Hypericum is not relevant, the comment is still useful - it answers the question as a knowledgeable contributor
- Comments are 3-6 paragraphs of flowing prose - no bullet points, no filler phrases

Layer 6: Aggregation and Opportunity Ranking

All signals are grouped by cluster. Each cluster is scored on four dimensions:

Weight	Dimension	What it measures
40%	Frequency	Volume of posts - proves the pain is recurring
30%	Intent / urgency	How intensely people are suffering
20%	Engagement	Upvotes and comment counts - measures community resonance
10%	Trend velocity	How many signals are recent - measures timing relevance

The result is a ranked list of the clusters generating the most opportunity right now. This tells you which problem domains are currently most active on Reddit.

Layer 7: Export and Review

Mod menu dump: A moderator menu action collects all signals, runs any missing insight and draft extractions (backfill), and prints the full output to the development terminal.

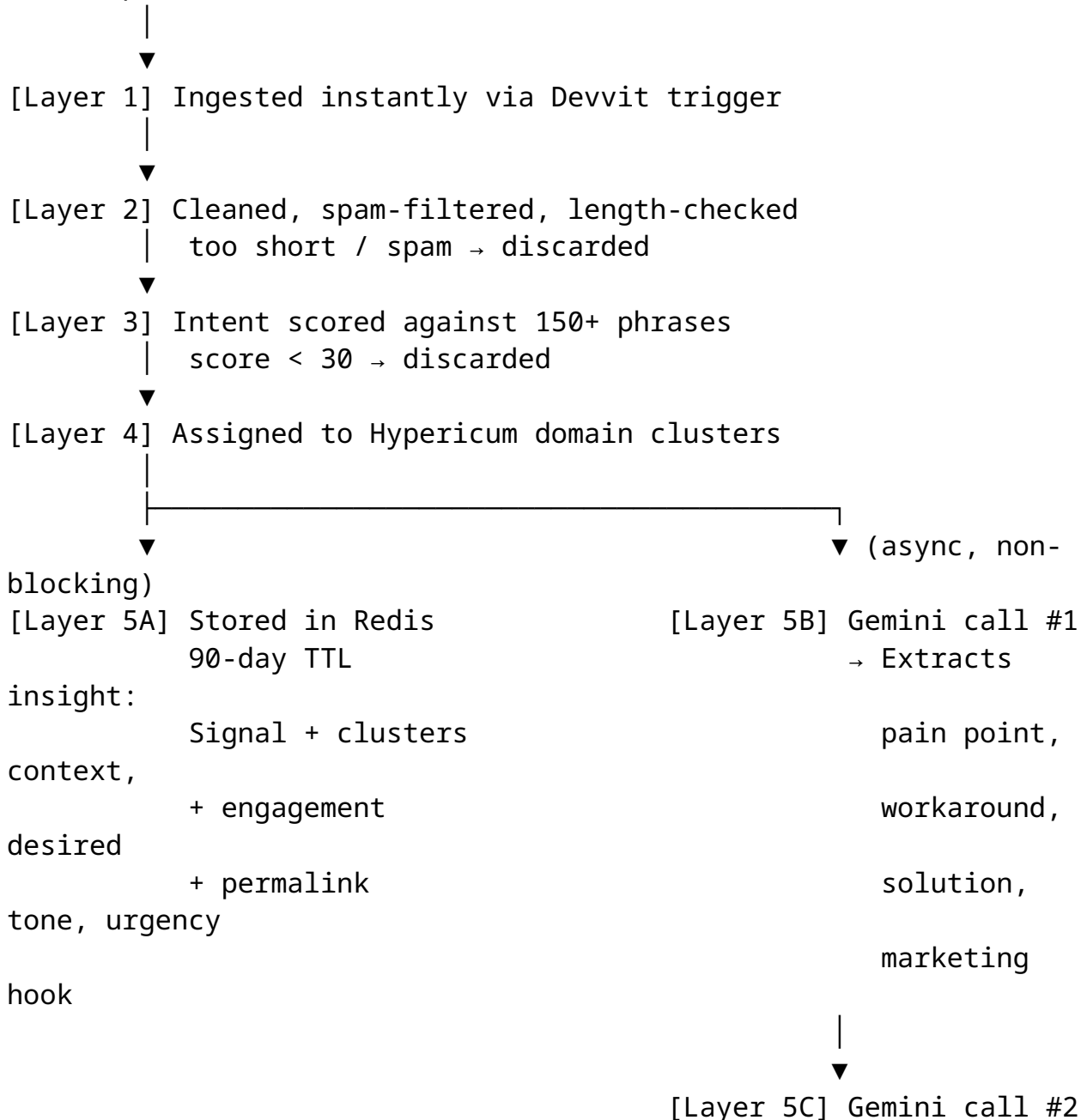
Local JSON export: - Every signal with its full metadata - The LLM insight card for each signal - The comment draft for each signal - The ranked opportunity list

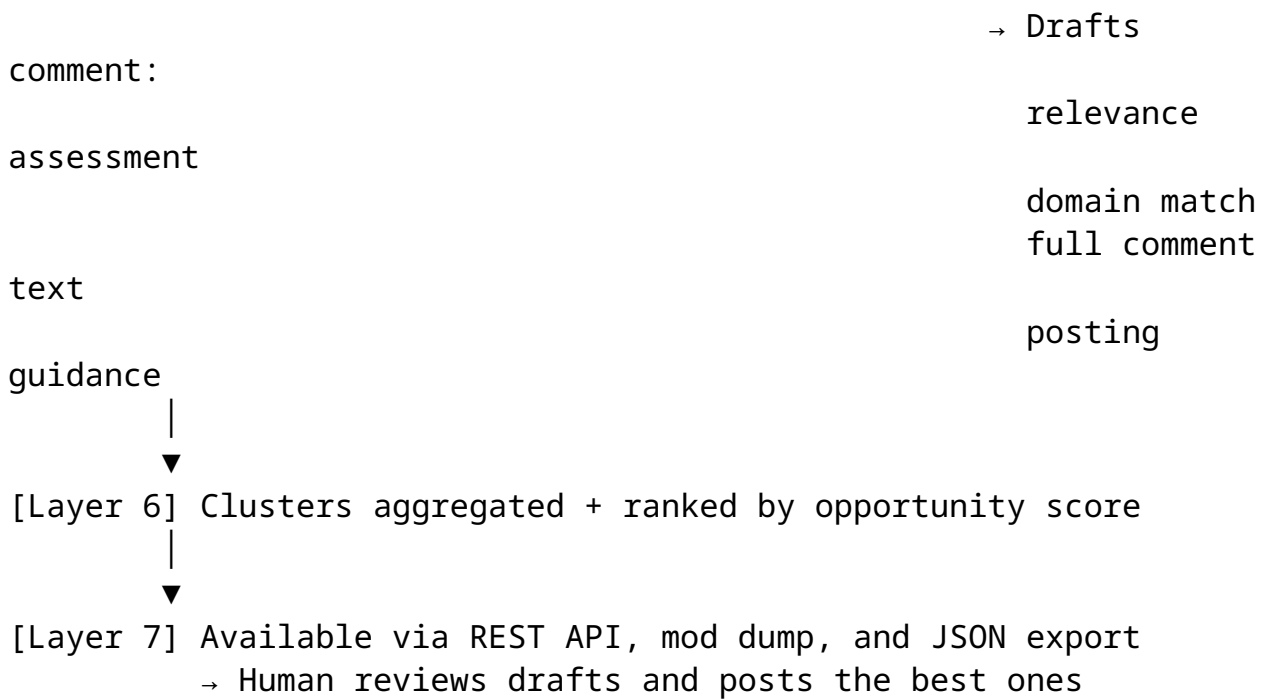
REST API: The system exposes live endpoints:

Endpoint	Returns
GET /api/signals	All raw signals
GET /api/aggregated-signals	Signals grouped by cluster
GET /api/ranked-opportunities	Clusters ranked by opportunity score
GET /api/comment-drafts	All drafts with signal and insight context
GET /api/comment-drafts?relevance=direct	Only posts where Hypericum is the direct answer
GET /api/config	Current subreddit configuration

The Full Pipeline in One View

Reddit post / comment submitted





What Has Been Built (Phase Status)

Phase	Description	Status
Phase 1	Event ingestion, Redis storage, subreddit config	Done
Phase 2	Preprocessing, intent scoring, clustering, aggregation, ranking, API	Done
Phase 3	LLM insight extraction (Gemini 2.5 Flash)	Done
Phase 4	Hypericum domain tuning + comment draft generation	Done

What Is Next

Phase 5-A: Deploy to real subreddits

Submit the Devvit app for Reddit’s app review process and install it on the subreddits where Hypericum’s target audience is active:

- r/dataengineering
- r/BusinessIntelligence
- r/MachineLearning
- r/SaaS
- r/devops

- r/entrepreneur
- r/dataanalysis
- r/Palantir (for competitive positioning)
- r/dbt (for positioning vs. metric governance tools)

This requires a Reddit account with moderator access to these subreddits, or working with subreddit moderators to install the app.

Phase 5-B: Native Reddit dashboard UI

A custom interface embedded directly in Reddit (built using Devvit's UI framework) showing live ranked opportunities and comment draft cards - no terminal required. The reviewer sees the post, the insight, the draft, and can copy it in one click.

Phase 5-C: Proactive subreddit search

The current system is passive - it only sees new content in subreddits where it's already installed. A proactive search layer would query Reddit's search API on a schedule for specific keywords matching Hypericum's six problem domains, expanding coverage beyond installed subreddits. This catches high-value historical posts and conversations in communities where Hypericum isn't yet a moderator.

Phase 5-D: Real-time alerting

A webhook or direct message notification fires whenever a high urgency signal is saved with direct relevance - meaning someone is actively and urgently describing an exact Hypericum problem right now. This ensures high-priority opportunities are reviewed within hours, not days.

Phase 5-E: Engagement analytics

Track which posts received a comment (posted externally by the team), and measure the response: upvotes, replies, profile visits. Closes the loop between signal detection and actual community impact.